DIPLOMA CURRICULUM REVISION 2026 · COURSE 2032

Electric Circuits and Systems

This course develops systematic DC and AC circuit analysis using Kirchhoff laws, mesh/nodal methods, network theorems, phasor algebra, resonance and balanced three-phase systems. Numerical problem-solving and phasor interpretation are central.

REV2026 · **Semester II** · **Theory + Tutorial** · **Malayalam support** · **Printable handbook**

Course code	**2032**
Semester	**II**
Course type	**Theory + Tutorial**
Contact hours	**60**
Credits	**4**
CIE / ESE	**40 / 60**
Modules	**4**

COURSE OVERVIEW

What this handbook covers

The structure follows the supplied official SITTTR Kerala Revision 2026 syllabus. It is a learning handbook rather than a replacement for the official syllabus PDF. Use the module checklist to ensure that every prescribed point has been revised.

CO1

Solve DC networks using Kirchhoff laws, mesh/nodal analysis, network theorems and star–delta transformation.

CO2

Apply phasor algebra to series R, L, C, RL, RC and RLC circuits and compute power quantities.

CO3

Analyse series/parallel resonance and two-branch parallel AC circuits.

CO4

Analyse balanced star/delta three-phase systems and compute line/phase quantities and power.

Module roadmap

Module I
Electric circuits and network theorems

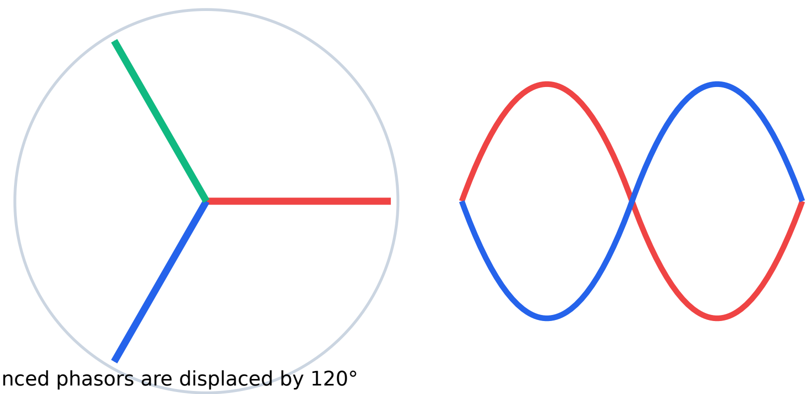
Module II
Phasor algebra and series AC circuits

Module III
Resonance and parallel AC circuits

Module IV
Balanced three-phase AC systems

VISUAL CONCEPT

See the subject as a system



Balanced phasors are displaced by 120°

MODULE I

15 (3.5 L + 11.5 T) CO1

Electric circuits and network theorems

KCL node-ൽ current conservation ആണ്; KVL closed loop-ൽ voltage conservation ആണ്. Sign convention ഒരിക്കൽ തിരഞ്ഞെടുക്കുക, equation മുഴുവൻ അതേ രീതിയിൽ പാലിക്കുക. Thevenin/Norton equivalent load മാറ്റുമ്പോൾ analysis വേഗത്തിലാക്കുന്നു.

Circuit types and terms

Identify active/passive, linear/nonlinear, bilateral/unilateral, lumped/distributed and independent/dependent-source circuits. Define branch, node, loop and mesh.

Kirchhoff current and voltage laws

Write algebraic current sum at a node and voltage sum around a loop. Establish current directions and voltage polarities before equations.

Mesh analysis

Assign mesh currents, write self and mutual resistance terms and solve up to three simultaneous equations using elimination, matrices or Cramer’s rule.

Nodal analysis

Choose a reference node, express branch currents in node voltages and solve up to three unknown node voltages.

Superposition theorem

In a linear network, consider one independent source at a time: ideal voltage sources become shorts and ideal current sources become opens. Power is not superposed directly.

Thevenin theorem

Remove the load, calculate open-circuit Vth and equivalent resistance Rth with independent sources deactivated, then reconnect the load.

Norton theorem

Find short-circuit current IN and equivalent RN = Rth. Convert between Thevenin and Norton forms using Vth = IN Rth.

Maximum power transfer

For DC resistive networks, maximum load power occurs at RL = Rth; efficiency is 50% at that condition.

Star–delta transformation

Convert three-terminal resistor networks without derivation and use the conversion to reduce otherwise non-series/parallel DC circuits.

Rules / common mistakes

- Mark assumed directions; a negative answer means opposite direction.
- Deactivate only independent sources when finding equivalent resistance.
- Never short an ideal voltage source in the original live circuit.
- Check power balance where practical.

Practice and self-learning

- Solve one three-mesh and one three-node network.
- Verify superposition against direct Kirchhoff solution.
- Solve the same load using Thevenin and Norton equivalents.
- Reduce a bridge section using star–delta conversion.

MODULE II

15 (10 L + 5 T) CO2

Phasor algebra and series AC circuits

Phasor sinusoidal steady-state quantity-യുടെ magnitude/phase representation ആണ്. R-ൽ voltage/current same phase; L-ൽ current 90° lag; C-ൽ current 90° lead. Complex-number sign തെറ്റിയാൽ power factor തെറ്റും.

j operator and complex plane

$j = \sqrt{-1}$ represents 90° rotation. Use rectangular $a + jb$ for addition/subtraction and polar $M\angle\theta$ for multiplication/division.

Rectangular–polar conversion

$M = \sqrt{a^2+b^2}$, $\theta = \text{atan2}(b,a)$; $a = M\cos\theta$ and $b = M\sin\theta$. Preserve the correct quadrant.

Pure resistance

$Z = R$, current and voltage are in phase, $P = VI$ and $Q = 0$.

Pure inductance

$X_L = \omega L$, $Z = jX_L$, current lags voltage by 90° and average active power is zero ideally.

Pure capacitance

$X_C = 1/(\omega C)$, $Z = -jX_C$, current leads voltage by 90° and average active power is zero ideally.

Series RL, RC and RLC

Add impedances as complex numbers. Draw phasor diagram and impedance triangle; calculate current, phase angle and power factor.

Power triangle

Active $P = VI\cos\varphi$, reactive $Q = VI\sin\varphi$ and apparent $S = VI$; $S^2 = P^2 + Q^2$. Identify leading or lagging power factor.

Series-circuit problems

Compute element voltages, total impedance, current and power. Verify that phasor voltages add to the supply.

Rules / common mistakes

- Use RMS values for AC power unless stated otherwise.
- Keep inductive reactance positive j and capacitive reactance negative j.
- Use atan2 or quadrant correction.

- State whether power factor is leading or lagging.

Practice and self-learning

- Convert five phasors between polar and rectangular form.
- Solve pure R/L/C and series RL/RC/RLC circuits.
- Draw impedance and power triangles from numerical results.

MODULE III

15 (7 L + 8 T) CO3

Resonance and parallel AC circuits

Series resonance-ൽ XL = XC, impedance minimum, current maximum. Parallel resonance-ൽ input current minimum/impedance maximum എന്നതാണ് സാധാരണ behaviour. Q factor selectivity കാണിക്കുന്നു; bandwidth = fr/Q.

Series resonance

For a series RLC circuit, resonance occurs when XL = XC and net reactance is zero. Derive fr = 1/(2π√LC).

Q factor

For series resonance Q = ωrL/R = 1/(ωrCR). Explain voltage magnification and selectivity.

Bandwidth

BW = f2 − f1 = fr/Q for the standard series response. Define half-power frequencies.

Admittance quantities

Y = 1/Z in siemens, G is conductance and B is susceptance. Inductive susceptance is negative and capacitive susceptance positive under the usual convention.

Two-branch parallel circuits

Solve by admittance: add branch conductance and susceptance, find total Y, invert for Z and calculate source current. Also solve by rectangular/vector current addition.

Parallel resonance

State the applicable resonant-frequency expression for the circuit form given and explain circulating branch currents, high impedance and low supply current.

Series versus parallel resonance

Compare input impedance, source current, magnification, typical response and applications.

Applications

Tuning, filters, oscillators, impedance selection and frequency-sensitive networks.

Rules / common mistakes

- Do not use the series-resonance formula blindly for every parallel circuit topology.
- Add admittances in parallel, not impedances.
- Keep susceptance signs consistent.
- Q and bandwidth relations depend on the stated circuit model.

Practice and self-learning

- Calculate fr, Q and BW for a series RLC circuit.
- Solve two parallel branches by admittance and vector-current methods and compare.
- Sketch current/impedance frequency responses.

MODULE IV

15 (9.5 L + 5.5 T) CO4

Balanced three-phase AC systems

Three-phase voltages 120° apart ആണ്. Star connection-ൽ VL = √3 Vph, IL = Iph. Delta connection-ൽ VL = Vph, IL = √3 Iph. Line/phase relation mix ചെയ്യുന്നത് ഏറ്റവും സാധാരണ mistake ആണ്.

Generation of three-phase voltage

Three identical windings displaced by 120 electrical degrees produce balanced phase voltages. Explain waveforms and phasor diagram.

Phase sequence

ABC and ACB sequences determine rotating-field direction. Interchanging any two lines reverses sequence.

Advantages of three phase

More constant power, economical conductor use, self-starting motors and compact machines for a given output.

Star connection

Common neutral point: VL = √3 Vph with 30° displacement; IL = Iph for a balanced load.

Delta connection

Closed loop: VL = Vph and IL = √3 Iph with 30° displacement.

Balanced-load problems

Calculate phase impedance, phase current, line current, line voltage and neutral current for balanced star/delta systems.

Three-phase power

Star–delta comparison

For balanced systems $P = \sqrt{3}\ V_L I_L \cos\varphi$, $Q = \sqrt{3}\ V_L I_L \sin\varphi$ and $S = \sqrt{3}\ V_L I_L$. These expressions apply to both star and delta when line quantities are used.

Compare insulation requirement, neutral availability, phase voltage/current and common applications.

Rules / common mistakes

- Write connection type before selecting formulas.
- Use line quantities consistently in three-phase power formulas.
- Balanced-load neutral current is zero in star.
- Phase sequence is not the same as phase angle magnitude.

Practice and self-learning

- Draw positive and negative phase-sequence phasors/waveforms.
- Solve balanced star and delta load problems.
- Calculate active, reactive and apparent three-phase power.

QUICK REVISION

Formula / rule bank

Rule / Formula 1

KCL: $\Sigma I = 0$ at a node

Rule / Formula 2

KVL: $\Sigma V = 0$ around a loop

Rule / Formula 3

V_{th} = open-circuit voltage; R_{th} = resistance seen at load terminals

Rule / Formula 4

I_N = short-circuit current; $R_N = R_{th}$

Rule / Formula 5

Maximum DC power: $R_L = R_{th}$

Rule / Formula 6

Star → Delta: $R_{ab} = R_a + R_b + (R_a R_b / R_c)$, cyclic

Rule / Formula 7

Delta → Star: $R_a = R_{ab} R_{ca} / (R_{ab} + R_{bc} + R_{ca})$

Rule / Formula 8

$Z_R = R$; $Z_L = j\omega L$; $Z_C = 1/(j\omega C) = -j/(\omega C)$

Rule / Formula 9

$Y = 1/Z = G + jB$

Rule / Formula 10

$P = VI\cos\varphi$; $Q = VI\sin\varphi$; $S = VI$

Rule / Formula 11

$f_r = 1/(2\pi\sqrt{LC})$

Rule / Formula 12

Series $Q = \omega_r L/R = 1/(\omega_r CR)$

Rule / Formula 13

$BW = f_r/Q$

Rule / Formula 14

Star: $V_L = \sqrt{3} V_{ph}$, $I_L = I_{ph}$

Rule / Formula 15

Delta: $V_L = V_{ph}$, $I_L = \sqrt{3} I_{ph}$

Rule / Formula 16

Three-phase $P=\sqrt{3}V_LI_L\cos\varphi$; $Q=\sqrt{3}V_LI_L\sin\varphi$; $S=\sqrt{3}V_LI_L$

Worked checkpoints

Worked checkpoint 1

A Thevenin source 12 V with R_{th} 6 Ω delivers maximum power to R_L = 6 Ω; $P_{max} = V_{th}^2/(4R_{th}) = 6$ W.

Worked checkpoint 2

For R = 8 Ω and X_L = 6 Ω in series, Z = 8+j6 = 10∠36.87° Ω and power factor = 0.8 lagging.

Worked checkpoint 3

L = 10 mH and C = 100 μF gives $f_r \approx 159.2$ Hz.

Worked checkpoint 4

A balanced star load on 400 V line supply has $V_{ph} = 400/\sqrt{3} \approx 231$ V.

APPLICATION

Practical, drawing and self-learning work

No.	Task	Required evidence
1	Three-mesh and nodal-analysis problem set.	Record procedure, observation/result and safety/quality checks.
2	Superposition verification.	Record procedure, observation/result and safety/quality checks.
3	Thevenin/Norton load analysis.	Record procedure, observation/result and safety/quality checks.
4	Maximum-power and star–delta problems.	Record procedure, observation/result and safety/quality checks.
5	Phasor conversion and operations.	Record procedure, observation/result and safety/quality checks.
6	Pure R/L/C and series AC problems.	Record procedure, observation/result and safety/quality checks.
7	Impedance and power triangles.	Record procedure, observation/result and safety/quality checks.
8	Series resonance calculations and response sketch.	Record procedure, observation/result and safety/quality checks.
9	Parallel two-branch circuits by two methods.	Record procedure, observation/result and safety/quality checks.
10	Parallel-resonance comparison.	Record procedure, observation/result and safety/quality checks.
11	Three-phase generation/sequence diagrams.	Record procedure, observation/result and safety/quality checks.
12	Balanced star/delta and three-phase power problems.	Record procedure, observation/result and safety/quality checks.

EXAM PREPARATION

Expected questions and model-answer guides

These are syllabus-aligned practice questions, not a claim about the exact end-semester paper.

2 marks

Q1. Define branch, node, loop and mesh.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q2. State and apply KCL and KVL.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q3. Explain mesh and nodal analysis for up to three unknowns.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q4. State limitations and procedure of superposition theorem.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q5. Derive/use Thevenin and Norton equivalents.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q6. State maximum power transfer theorem and solve a problem.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q7. Convert star to delta and delta to star.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q8. Explain j operator and phasor conversion.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q9. Analyse pure R, L and C under AC.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q10. Solve a series RLC circuit and draw impedance/power triangles.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q11. Derive resonant frequency of series RLC.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q12. Define Q factor and bandwidth.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q13. Solve a two-branch parallel AC circuit using admittance.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q14. Compare series and parallel resonance.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q15. Explain generation and phase sequence of three-phase voltage.

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q16. Derive line/phase relations in star and delta.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8–10 marks

Q17. Calculate balanced three-phase active, reactive and apparent power.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

COVERAGE AUDIT

Official syllabus coverage checklist

- Module I: Circuit types and terms
- Module I: Kirchhoff current and voltage laws
- Module I: Mesh analysis
- Module I: Nodal analysis
- Module I: Superposition theorem
- Module I: Thevenin theorem
- Module I: Norton theorem
- Module I: Maximum power transfer
- Module I: Star–delta transformation
- Module II: j operator and complex plane
- Module II: Rectangular–polar conversion
- Module II: Pure resistance
- Module II: Pure inductance
- Module II: Pure capacitance
- Module II: Series RL, RC and RLC
- Module II: Power triangle
- Module II: Series-circuit problems
- Module III: Series resonance
- Module III: Q factor
- Module III: Bandwidth

- Module III: Admittance quantities

- Module III: Two-branch parallel circuits
- Module III: Parallel resonance
- Module III: Series versus parallel resonance
- Module III: Applications
- Module IV: Generation of three-phase voltage
- Module IV: Phase sequence
- Module IV: Advantages of three phase
- Module IV: Star connection
- Module IV: Delta connection
- Module IV: Balanced-load problems
- Module IV: Three-phase power
- Module IV: Star–delta comparison

REFERENCES

Learning resources

1. A Textbook of Electrical Technology Vol. I — B. L. Theraja & A. K. Theraja
2. Principles of Electrical Engineering — V. K. Mehta & Rohit Mehta
3. Basic Electrical Engineering — V. Mittal & Arvind Mittal
4. Fundamentals of Electrical Network — B. R. Gupta & Vandana Singhal
5. Circuit Theory: Analysis and Synthesis — Abhijit Chakrabarti

Primary authority for scope: supplied official SITTTR Kerala Diploma Curriculum Revision 2026 syllabus PDF for Course 2032.